TT-T-001386 (GSA-FSS)
June 9, 1977
SUPERSEDING
(See 6.3)

INTERIM FEDERAL SPECIFICATION

THINNER, SYNTHETIC RESIN ENAMEL AND PRIMER (FOR LOW AND HIGH TEMPERATURE SPRAY APPLICATIONS)

This Interim Federal Specification was developed by the General Services Administration, Federal Supply Service, Washington, DC 20406, based upon currently available technical information. It is recommended that Federal agencies use it in procurement and forward recommendations for changes to the preparing activity at the address shown above.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers two types of thinner for reducing synthetic resin equipment enamel and primer to the consistency suitable for fast dry spray application.

1.2 Classification. The thinner shall be of the following types:

Type I - For spray applications below 27°C (80°F).
Type II - For spray applications at 27°C (80°F) or higher.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

TT-E-001384 - Enamel, Gloss, Synthetic, for Metal Equipment.
TT-P-001385 - Primer Coating, Synthetic, Gray (For Metal Equipment).
PPP-P-1892 - Paint, Varnish, Lacquer, and Related Materials, Packing, and Marking of.

Federal Standard:

Fed. Test Method Std. No. 141 - Paint, Varnish, Lacquer, and Related Materials, Methods of Inspection, Sampling, and Testing.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

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Military Standard:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Chemical Society (ACS) Standards:

Reagent Chemicals (Fifth Edition, 1974).

(Application for copies should be addressed to the American Chemical Society, 1155 16th Street, NW, Washington, DC 20036).

American Society for Testing and Materials (ASTM) Standards:

D 1209 - Color of Clear Liquids (Platinum-Cobalt Scale).
D 3539 - Evaporation Rates of Volatile Liquids.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

3. REQUIREMENTS

3.1 Materials. The thinners shall consist of the materials specified in table I.

TABLE I. Formulas

Ingredient	Type I Percent by weight Min.	Type I Percent by weight Max.	Type II Percent by weight Min.	Type II Percent by weight Max.
Acetone	24.0	26.0	24.0	26.0
VM & P Naphtha*	40.0	42.0	40.0	42.0
Ethylene glycol monoethyl ether	8.5	9.5	----	----
Ethylene glycol monobutyl ether	----	----	8.5	9.5
Xylene	6.5	7.5	6.5	7.5
Toluene	10.5	11.5	10.5	11.5
Isobutyl acetate	6.5	7.5	6.5	7.5

*Less than 7 percent aromatic content.

3.2 Infrared spectrophotometric properties. When tested as specified in 4.3.2, band positions, shapes, and related intensities shall match the spectrum in figure 1 for the type I thinner, and figure 2 for the type II thinner.

3.3 Air pollution control. The thinners, when tested as specified in 4.3.3, shall conform by volume to the requirements controlling the emission of solvents into the atmosphere as called out in (a) through (f):

(a) Total of aldehydes and branched chain ketones: 20 percent maximum.
(b) Total of aromatic compounds with eight or more carbon atoms to the molecule except ethylbenzene: 8 percent maximum.
(c) Total of ethylbenzene and toluene: 20 percent maximum.
(d) Total of solvents with olefinic or cyclo-olefinic unsaturation: 5 percent maximum.
(e) Total of (a) + (b) + (c) + (d) = 20 percent maximum.
(f) Benzene: 0.4 percent maximum.

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3.4 Color. When tested as specified in table II, the thinners shall have a color not darker than the No. 10 standard.

3.5 Nonvolatile residue. When tested as specified in 4.3.4, 100 ml of each type thinner shall contain not more than 10 mg of nonvolatile residue.

3.6 Stain. When tested as specified in table II, the thinners shall show no evidence of stain or residual oiliness.

3.7 Evaporation rate. When tested as specified in table II, the thinners shall have the following evaporation rates:

Type I - Not less than 1.7 and not more than 1.9.
Type II - Not less than 1.5 and not more than 1.7.

3.8 Compatibility. When tested as specified in 4.3.5, the thinners shall not produce any coagulation, separation, precipitation, or other symptoms of incompatibility with enamel conforming to TT-E-001384 or with primer conforming to TT-P-001385.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Sampling and inspection.

4.2.1 Inspection of preparation for delivery. The packaging, packing, and marking shall be inspected in accordance with the requirements of PPP-P-1892 to determine conformance with section 5.

4.2.2 Sampling of the end item for tests. For end item testing, the filled unit containers of the thinner shall be assembled into lots, expressed in units of gallons, as specified in MIL-STD-105. In MIL-STD-105, the definition, "essentially the same condition", shall be interpreted to mean all the filled unit containers offered for delivery at one time. Samples from the lots shall be taken in accordance with MIL-STD-105, using inspection level S-2 and an acceptable quality level (AQL) of 2.5 percent defective.

4.3 Test procedures.

4.3.1 The thinner shall be tested in accordance with methods specified in table II, and as hereinafter specified. Unless otherwise specified, all tests shall be conducted at standard conditions which are 23° ± 1°C (74° ± 2°F) and 50 percent relative humidity. All test reports shall contain the individual values used in expressing the final result. All tests shall be evaluated for conformance with requirements in section 3. Failure to pass any test, or noncompliance with any requirement, shall be cause for rejection of the sample.

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TABLE II. Index of tests

Test	Method	Test reference	Requirement reference
Infrared spectroscopy	----------	4.3.2	3.2
Composition	----------	4.3.3	3.3
Aromatic & oxygenated compounds	7356 1/	4.3.3.1	3.3
Benzene	7356 1/	4.3.3.2	3.3
Color	D 1209 2/	-------	3.4
Nonvolatile residue	----------	4.3.4	3.5
Stain	4491 1/	-------	3.6
Evaporation rate	D 3539 2/	-------	3.7
Compatibility	----------	4.3.5	3.8

1/ Fed. Test Method Std. No. 141.
2/ American Society for Testing and Materials (ASTM).

4.3.2 Infrared spectrophotometric properties. Test the thinner for conformance with 3.2 in a sealed cell having a pathlength such that the strongest band is at about 5 percent transmission.

4.3.3 Thinner composition. Test the thinners for compliance with 3.3 in accordance with method 7356 of Fed. Test Method Std. No. 141, and as further specified in 4.3.3.1 and 4.3.3.2.

4.3.3.1 Aromatic and oxygenated thinners. The 1.83 m column shall be installed and the operating conditions described in method 7356 shall be followed. About 3 ml of the thinner shall be injected and the chromatogram scanned. The aliphatic solvents will emerge within 1 minute and the complete chromatogram should develop in about 5 minutes. From the position of the peaks observed on the chromatogram, an internal standard that will be free of interference shall be selected, such as cyclopentanol or cyclohexanol. Six-tenths of a ml of internal standard shall be added to 3 ml of the thinner. The sample shall be analyzed according to the above procedure. Peaks emerging after 1 minute are aromatic compounds along with any oxygenated compounds that may be present. The percent of aromatic and oxygenated compounds shall be calculated as follows:

$$\text{Percent aromatic and oxygenated compounds, v/v} = \frac{A \times B}{C \times D}$$

Where: A = percent of internal standard added (in this case, 20).
B = area of aromatic and oxygenated compounds.
C = calibration factor for the internal standard. This factor is dependent on the internal standard used and on the performance of the chromatograph, and should be determined daily.
D = area of the internal standard (in this case, cyclopentanol or cyclohexanol).

4.3.3.2 Benzene. When the thinner is tested in accordance with 4.3.3.1, a trace benzene peak of not more than 2 percent of the toluene peak will be allowed.

4.3.4 Nonvolatile residue. Place 100 ml of thinner in a tared porcelain dish. Evaporate the contents over a steam bath almost to dryness. Then heat the dish and residue to constant weight in an oven at 105 ± 2°C. Determine the weight of the nonvolatile residue in the dish. Conduct the test in duplicate and report the average result in g/100 ml for conformance with 3.5.

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4.3.5 Compatibility. Pour 100 ml of each of the following into separate 250 ml high-form glass beakers: Synthetic primer conforming to TT-P-001385, and one of the synthetic enamels conforming to TT-E-001384. To each beaker add 100 ml of the thinner, and stir with a glass rod without making bubbles. Immediately flow out a thin film of each mixture on separate clear glass plates. Examine the films for such nonuniformities as small gelled particles, which will indicate coagulation of the enamel or primer vehicle and constitute failure of the requirements in 3.8. Small gelled particles may be differentiated from possible pigment flocculation by probing with a sharp-pointed metal object. After flowing out the films, allow the mixtures in the beakers to remain undisturbed for 2 hours, and then examine the settled portion for possible coagulation of the primer or enamel vehicle. Any settling which is not soft and easily redispersed by light stirring with a glass rod shall indicate failure to meet the requirements of 3.8.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing, and marking. When specified, the thinners shall be packaged, packed, and marked in accordance with PPP-P-1892. The level of packaging and the level of packing shall be A or Commercial, as specified. The thinners shall be furnished in 1-gallon rectilinear cans, 5-gallon pails with incorporated pouring device, and 55-gallon drums. Unless otherwise specified, commercial packaging and packing shall be used (see 6.2).

5.2 Special markings. Special markings on all unit containers shall be as follows:

(a) "Warning! Flammable. Keep away from sparks and open flames."
(b) "Use with adequate ventilation."
(c) "Avoid prolonged breathing of vapors."

6. NOTES

6.1 Intended use. These thinners are intended for use in reducing the viscosity of synthetic resin enamels and paints in such manner as to facilitate application, especially by spraying. The type I thinner is intended for spray applications below 27°C (80°F). The type II thinner is intended for spray applications at 27°C (80°F) or higher.

6.2 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Type of thinner required (see 1.2).
(c) Size and type of container required (see 5.1).
(d) Level of packaging and packing required (see 5.1).
(e) Special markings required (see 5.2).

6.3 Supersession data. This specification supersedes General Services Administration purchase descriptions as follows:

TT-T-001386	GSA Purchase Description
Type I	PD-FSS-6A
Type II	PD-FSS-7A

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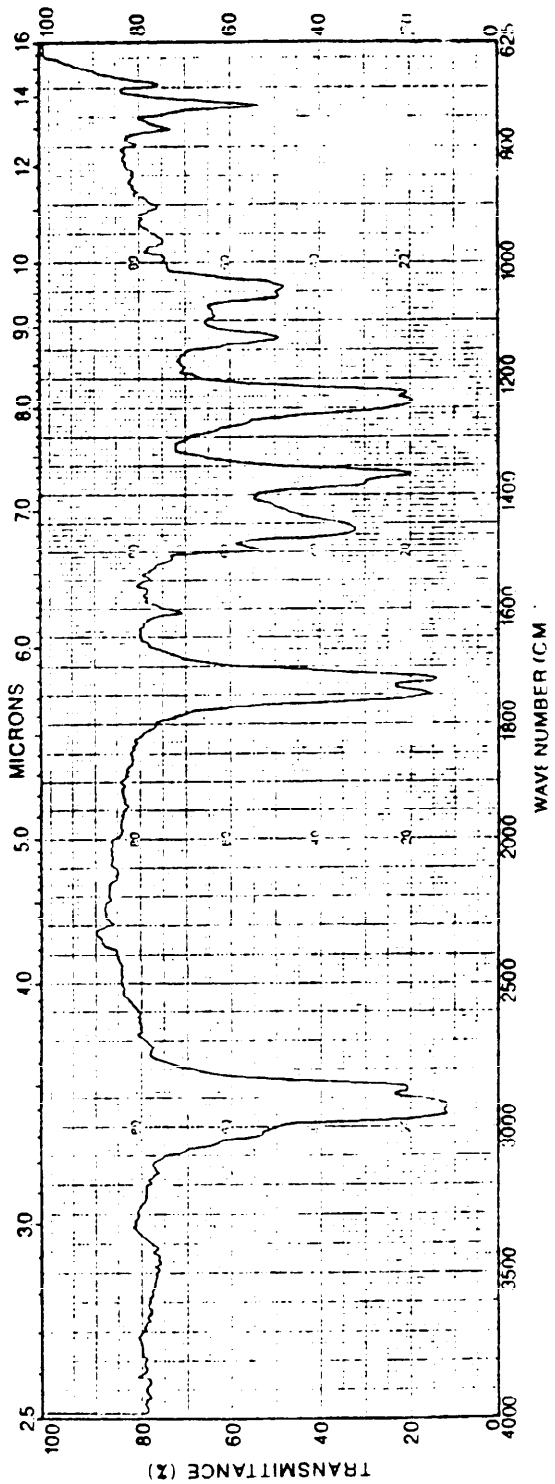


FIGURE 1 - Infrared Spectrophotometric Analysis of Type I Thinner

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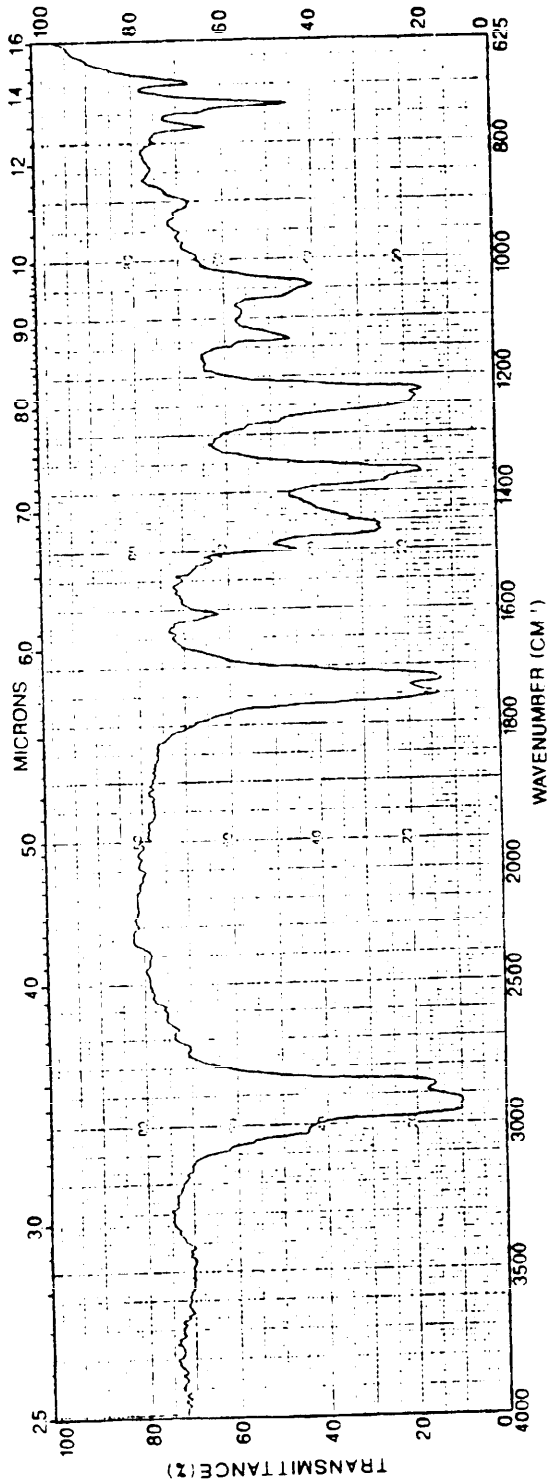


FIGURE 2 - Infrared Spectrophotometric Analysis of Type II Thinner

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